Bug Report Classifier - Replication

Jack Honour [3063159]

Replication

See `requirements.pdf` before running anything to ensure you have the necessary hardware and software requirements. As well as a correct virtual environment setup with the required dependencies installed.

```
(venv) python analysis.py
```

This will run the full analysis, replicating the results for all algorithms across all projects. Note that the improved_6 algorithm (BERT-based) is very time-consuming to train, so by default it will only run one repetition for each project.

The replicated results will be saved in the `results_replicated` folder, with the same structure as the original `results` folder. Each algorithm will have its own subfolder, and within each subfolder there will be CSV files for each project and repetition.

Custom Replication

The following arguments can be passed to `analysis.py` to customise the replication process:

- `--algorithm`: Specify which algorithm(s) to replicate. Can be one or more of `baseline`, `improved_1`, `improved_2`, `improved_3`, `improved_4`, `improved_5`, `improved_6`, `final`, `*`. Default is * algorithms.
- `--project`: Specify which project(s) to replicate. Can be one or more of `caffe`, `incubator-mxnet`, `keras`, `pytorch`, `tensorflow`, `all`, `*`. Default is * projects.
- `--repetitions`: Specify how many repetitions to run for each algorithm and project. Default is 10, 20 and 50 repetitions combined (except for improved_6 which defaults to 1 repetition due to time constraints).
- `--use-cached-models`: Whether to use cached pre-trained models if available, or always train from scratch. Default is to use cached models if available. Can be set to `true` or `false`.

Example, to replicate only the final algorithm all individual projects with 2 repetitions, you would run:

```
(venv) python analysis.py --algorithm=final --project=* --repetitions=2
```

Not to be confused with the all-projects analysis, which runs each algorithm on a combined dataset of all projects. To replicate the all-projects analysis for the final algorithm with 5 repetitions, you would run:

```
(venv) python analysis.py --algorithm=final --project=all --repetitions=5
```